INCH-POUND

MIL-PRF-1/1631B
9 July 1998
SUPERSEDING
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26 March 1973

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON
TYPE VA-220B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Reflex, tunable, waveguide output.

ABSOLUTE RATINGS: 1/

Parameter:	Ef	Ers	Er	F	Body temp	Ehk
Unit:	V	V dc	V dc	GHz	°C	V dc
Maximum:	7.0	775	-1,000	7.425	150	±45
Minimum:	5.7	----	2/	7.125	----	----

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1

TEST CONDITONS: 3/

Parameter:	Ef	Ers	Er	F	Body temp
Unit:	V	V dc	V dc	GHz	°C
	6.3	750	-250 to 400	(see tunable freq)	100 (max)

See footnotes at end of table I.

GENERAL:

Qualification - Required. 13/

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TABLE I. Testing and inspection.

Inspection	Method	Conditions	Symbol	Limits		Unit
				Min	Max	
Conformance inspection, part 1						
Emission	1231	Er = -350 to -450 V dc; Ef = 5.7 V 6/ 7/	ΔIrs/Irs	----	-15	%
Resonator current	4212	Er/max Po	Irs	----	80	mA dc
Reflector voltage	4213	Er/max Po 2/	Er	-250	-400	V dc
Total reflector current	4229	Er = -350 to -450 V dc 7/	Ir	----	5	μA dc
Heater current	1301	Ef = 6.3 V	If	----	0.9	A
Mechanical tuning	4223	Er/max Po 9/	F	7.125	7.425	GHz
Power output	4211	Er/max Po	Po	1.0	----	W
Electronic tuning range	4201	Er/50 percent max Po	ΔF	28	----	MHz
Hysteresis	4231	Er/max Po 8/	Ratio	----	50	%
Operating torque	4223		Torque	----	32	oz.-in.
Conformance inspection, part 2						
Heater frequency modulation	----	Ef = 6.3 V at 400 Hz	ΔF	----	30	kHz (p-p)
Temperature coefficient	4027	Er/max Po; TA = +25° to +80° 4/	ΔF/ΔTA	----	±100	kHz/°C
Reflector voltage modulation coefficient	4215	Er/max Po; ΔF < 5 MHz	ΔF/ΔEr	225	525	kHz/V
Beam voltage modulation coefficient	----	Er/max Po; ΔErs = +9 V dc	ΔF/ΔErs	----	±150	kHz/V
Conformance inspection, part 3						
Service-life guarantee	----	Er/max Po	t	7,500	----	hr
Service-life guarantee end points:	----	Er/max Po	Po	0.80	----	W
Periodic-check tests		10/				
Effective position of resonator	----	No voltages 5/	----	0.106	0.156	in.

1/ Ratings should not be exceeded under continuous or transient conditions. A single rating may be the limitation, and simultaneous operation at another rating may not be possible. Design values for systems should include a safety factor to maintain operation within ratings under voltage and environmental variations.

2/ Reflector voltage must always be negative with respect to the cathode by at least 20 volts.

3/ All oscillation tests shall be made with the tube rigidly connected to a UG-344/U (MIL-F-3922/55) flange on RG-50/U (MIL-W-85/1) waveguide equipment. The load VSWR for the tube shall be less than 1.1:1. Forced-air cooling of at least 30 cfm is required in the fin and the waveguide as shown in figure 1. For maximum tube life, the tube temperature should be less than 100°C.

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TABLE I. Testing and inspection - Continued.

4/ The temperature coefficient, when tested under the conditions specified, shall be computed in kilohertz per degree celsius over the temperature range specified.

5/ The tube under test (TUT) shall be adjusted to its center (resonance) frequency. Tune the signal source -150 MHz from the tube's center frequency and locate the position of the null on a standing wave indicator. Repeat the procedure with the signal source tuned to +150 MHz from the tube's center frequency. Replace the TUT with a shorting plate, tune the signal source to the tube's center frequency, and locate the position of the null on the standing-wave indicator. The effective resonator position is then defined as the position of the VSWR null with the shorting plate attached to the waveguide minus the average location of the VSWR null at ±150 MHz from the tube's center frequency with the tube attached to the waveguide. The VS-1120 adapter, or equal, may be used to change the effective resonator position of the tube 0.710 ± 0.005 inches.

6/ The beam current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.

7/ The test shall be made after a minimum warm-up of 2 minutes. The tube shall not be oscillating during the test.

8/ There shall be no power discontinuities between the half-power points for either direction of reflector voltage change through the tuning range specified under tunable frequency.

9/ The tube shall be capable of being tuned smoothly and without discontinuities over a frequency range including the limits specified.

10/ If the periodic-check tests have previously been performed by the tube manufacturer, and if materials, manufacturing processes, and test requirements are the same as when the tests were previously performed, the periodic-check test(s) need not be performed.

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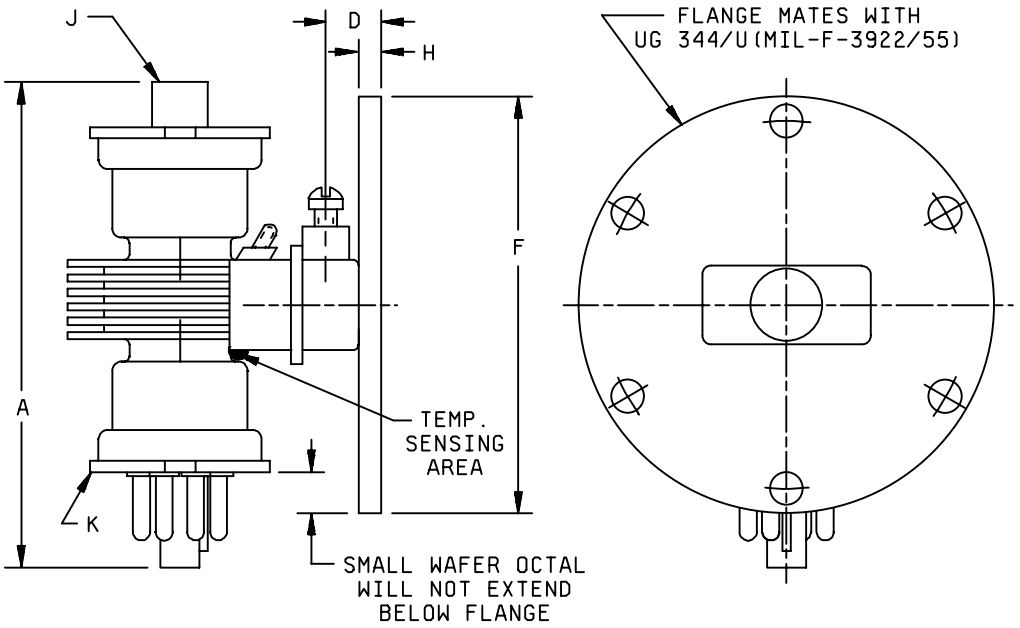

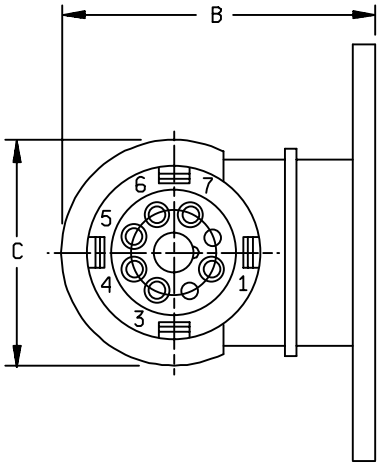

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 1				
A	----	3.750	----	95.25
B	----	2.375	----	60.33
F	----	3.190	----	81.03
J	CAP: C1-1			
K	BASE: B6-242			
Conformance inspection, part 2				
C	----	1.750	----	44.45
D	0.365	0.395	9.27	10.03
H	0.105	0.135	2.67	3.43

Pin Connections	
No.	Element
1	Shell
3	No Connection
4	Cathode
5	Heater
6	Heater
7	No Connection
Cap	Reflector

NOTES: Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type VA-220B.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11, 17, 19, 99

Preparing activity:
DLA - CC

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